Lecture 7

Realtime PCR, Nested PCR, Multiplex PCR, and Gap PCR

Instructor: Dr. Luu Phuc Loi

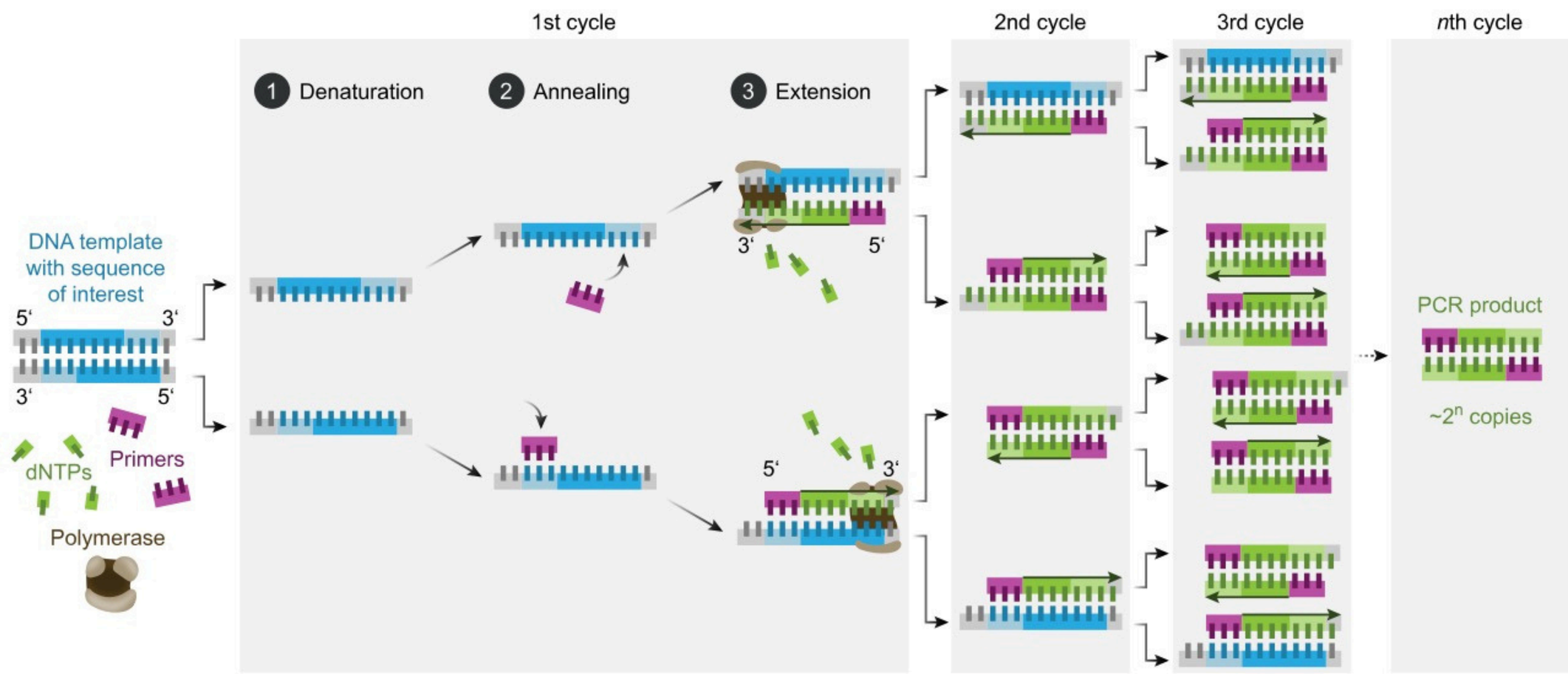
Teaching Assistant: Hoang Nhu Quynh

Date: Nov 13, 2026

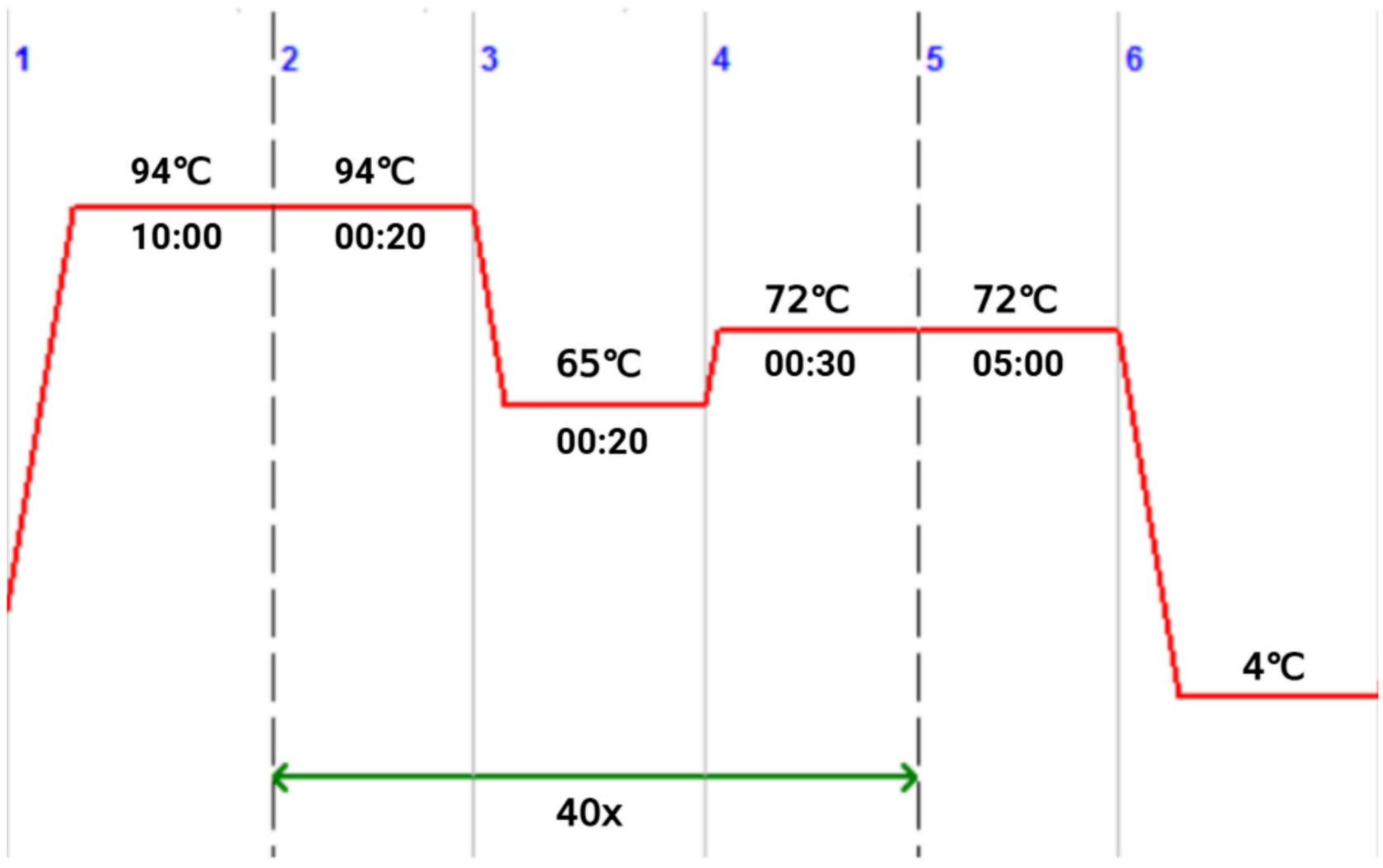
WHAT IS PCR ??

The polymerase chain reaction used for amplification of a specific DNA region

- Two primers
- DNA polymerase
- Repeated heating and cooling cycles



PCR Steps

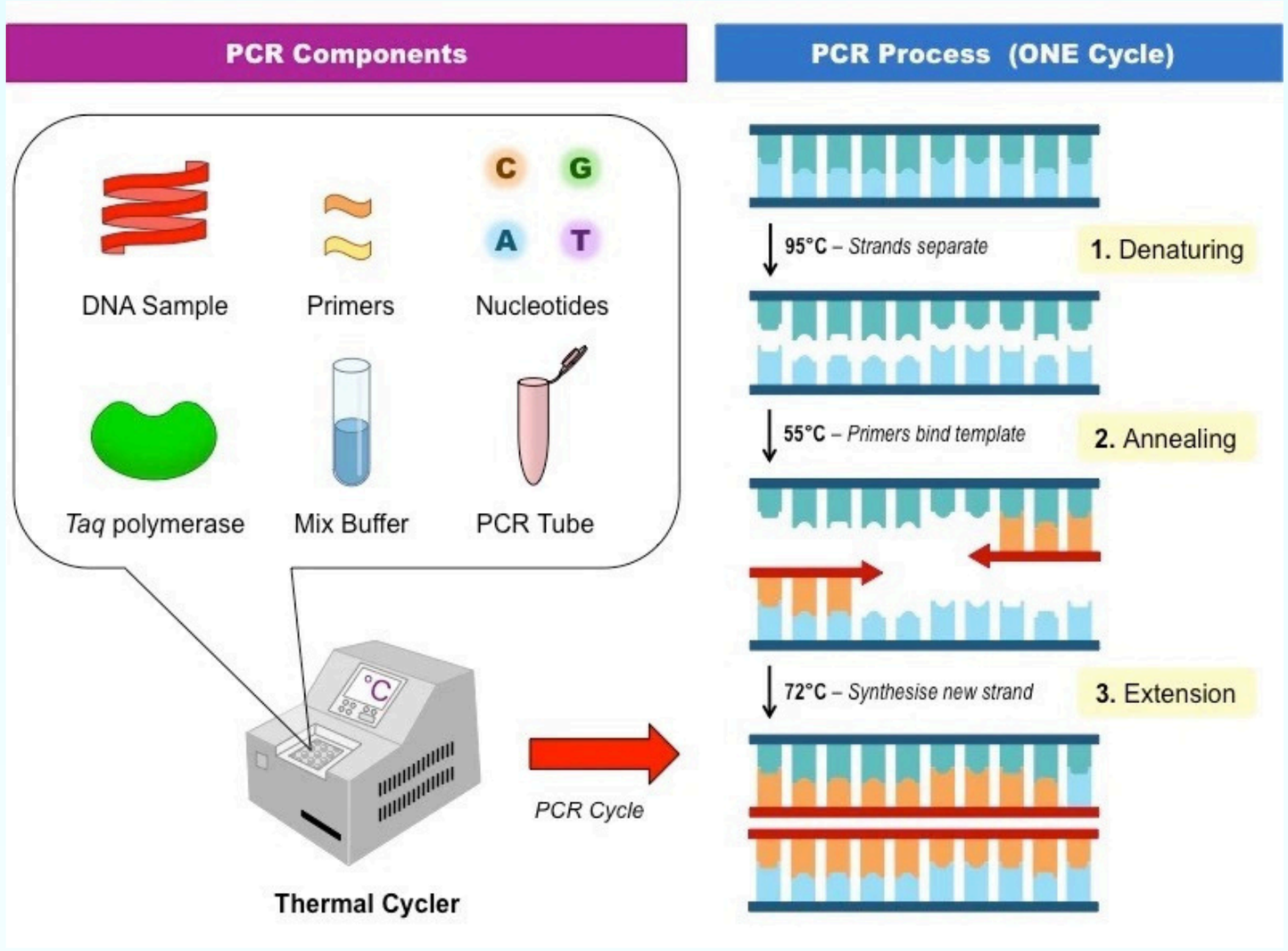


PCR Components

Note:

Mix Buffer

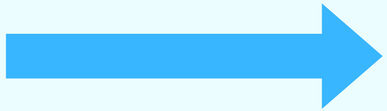
- Mg2+ (MgCl2)
- PCR Buffer



What is a PCR control ??

Used to determine the accuracy and reproducibility of the experiment.
=> increase the trustworthiness of the exp.

- Negative Control => no template
- Positive Control => its product is same size compared to target gene
- Internal Control



What PCR is used for ??

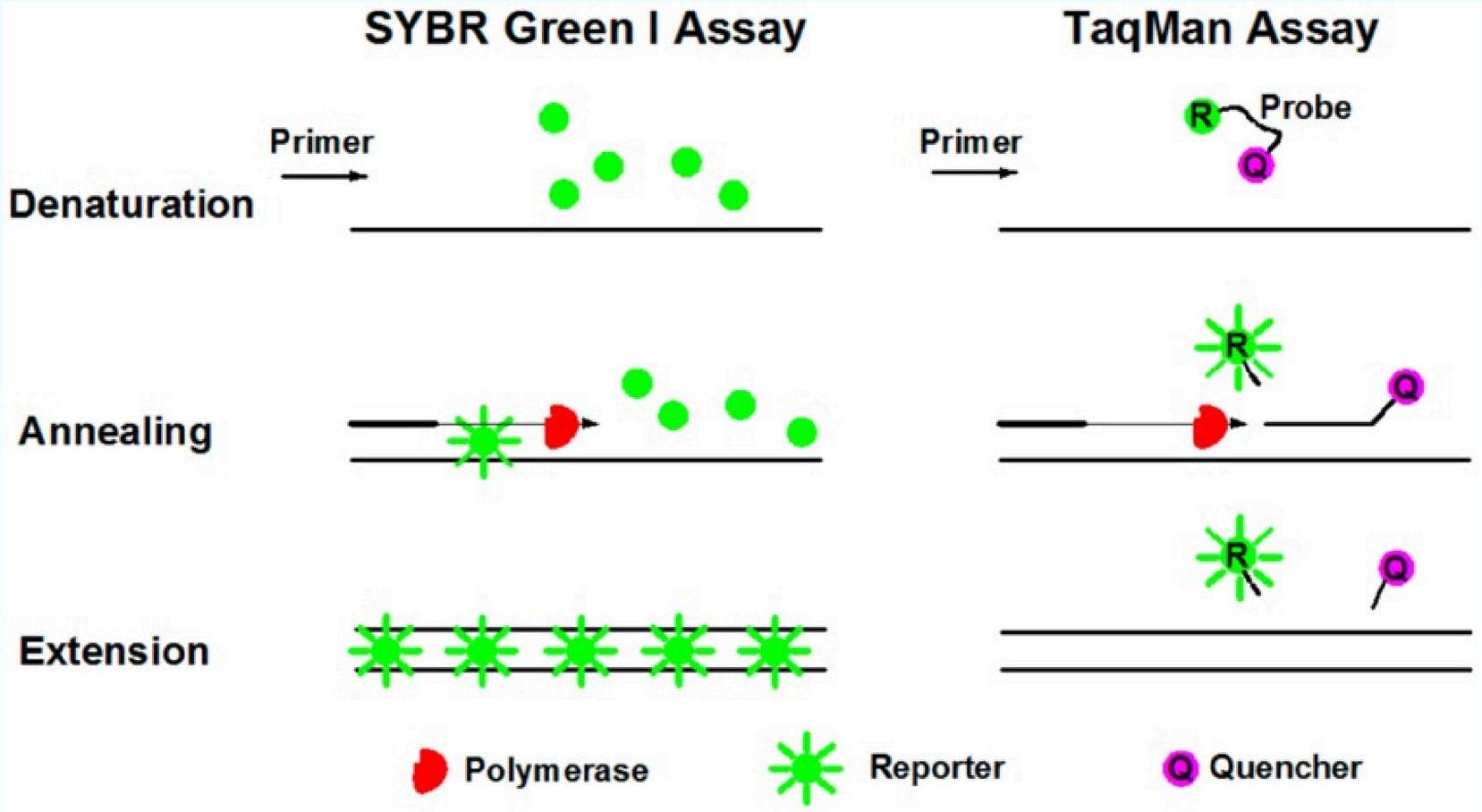
Limits of PCR

REALTIME PCR

Also known as quantitative PCR (qPRC), measures fluorescence at every cycle and quantifies during exponential phase

Fluorescence Dye

- SYBR Green, detects any double-stranded DNA (dsDNA) produced during PCR
- TaqMan, uses a sequence-specific fluorescent probe to detect the target DNA

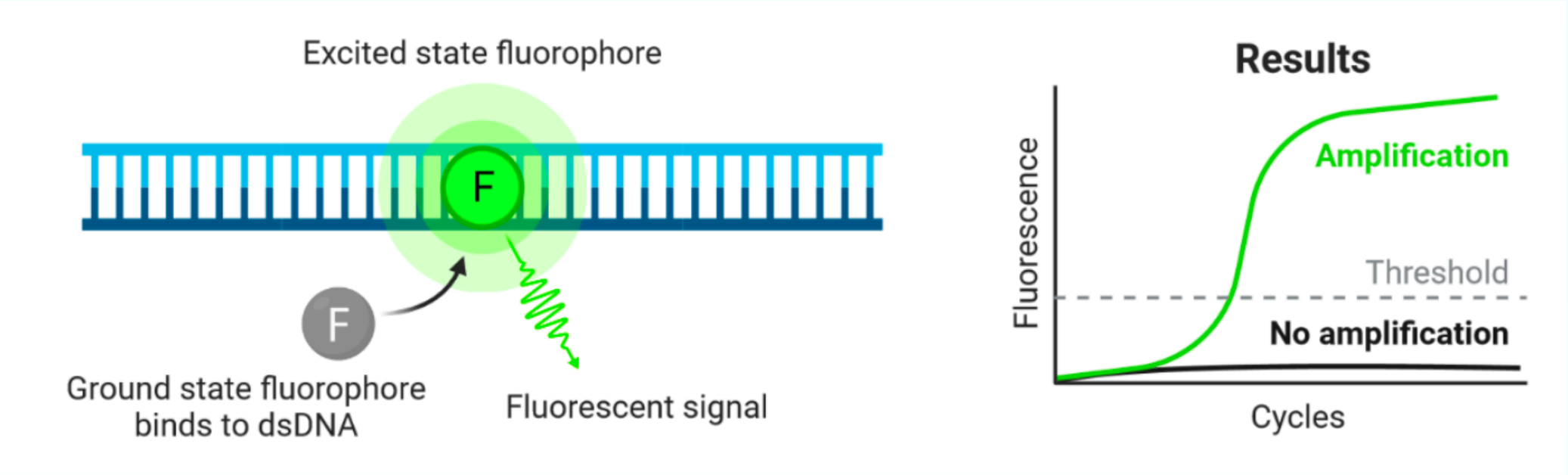


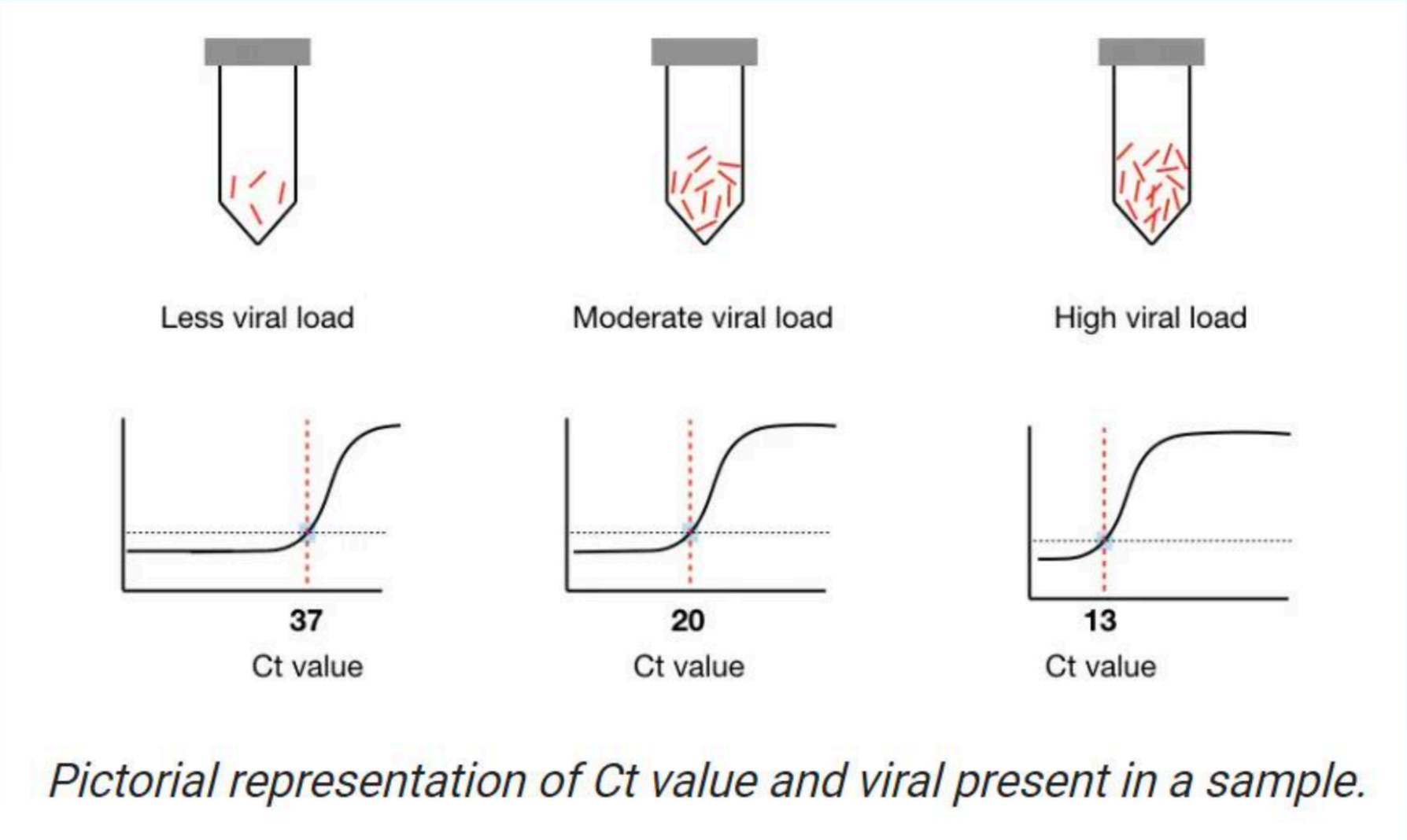
Key concept

The higher concentration of dsDNA => the higher fluorescent intensity signal.

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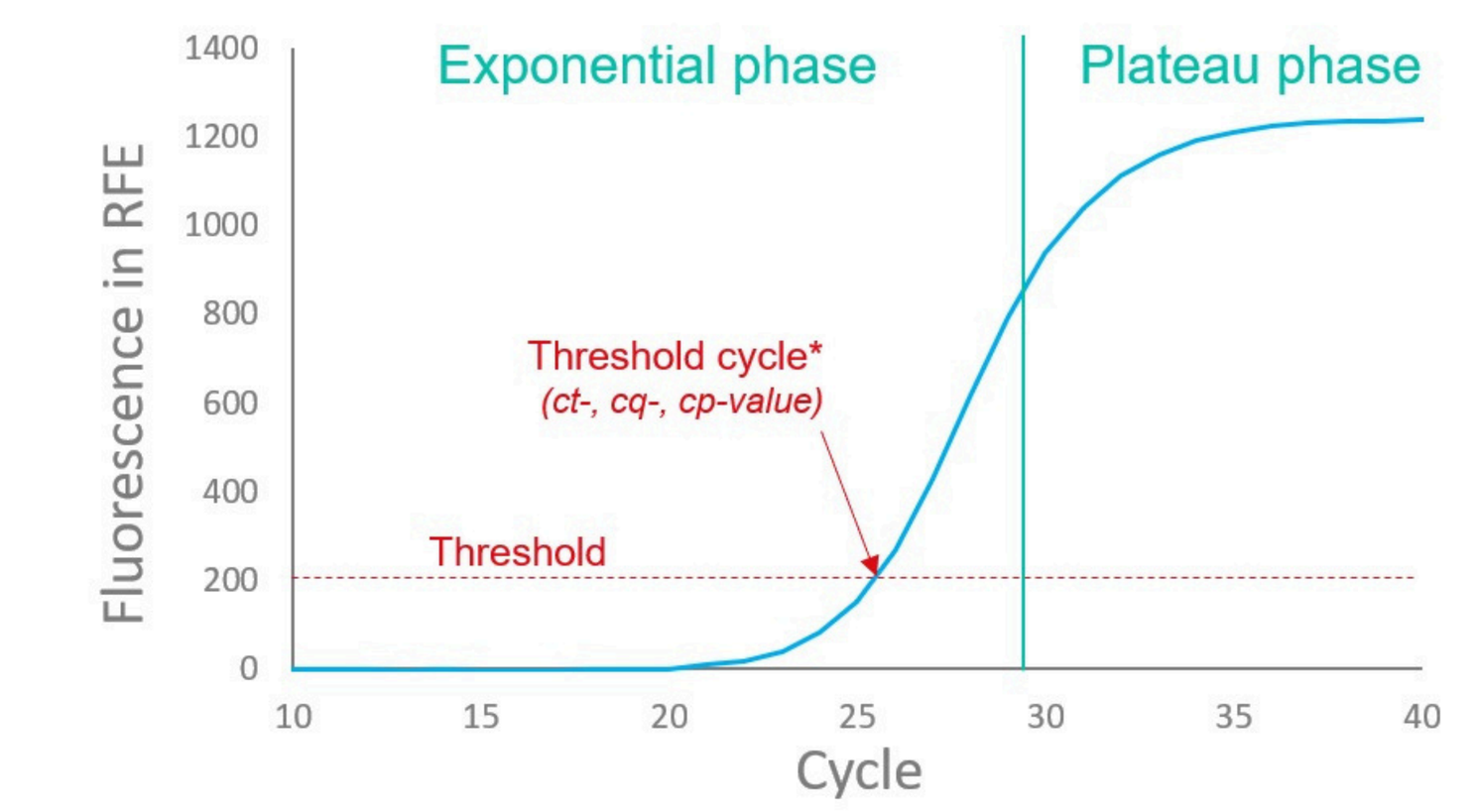


Pictorial representation of Ct value and viral present in a sample.

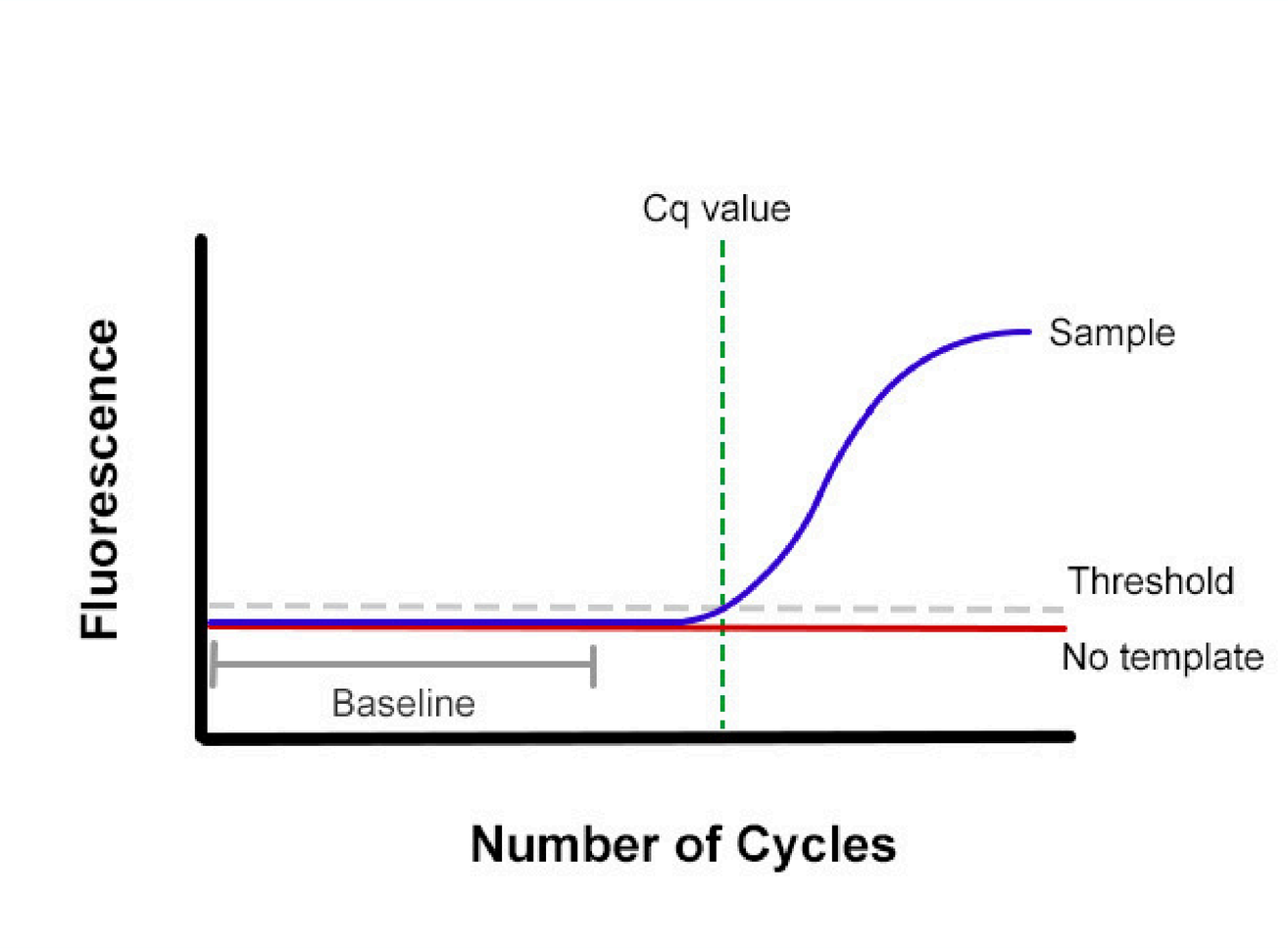
(Source: https://geneticeducation.co.in/what-is-the-ct-value-of-sars-cov-2-covid-19-rt-pcr/#google_vignette)

Realtime PCR kinetics phases

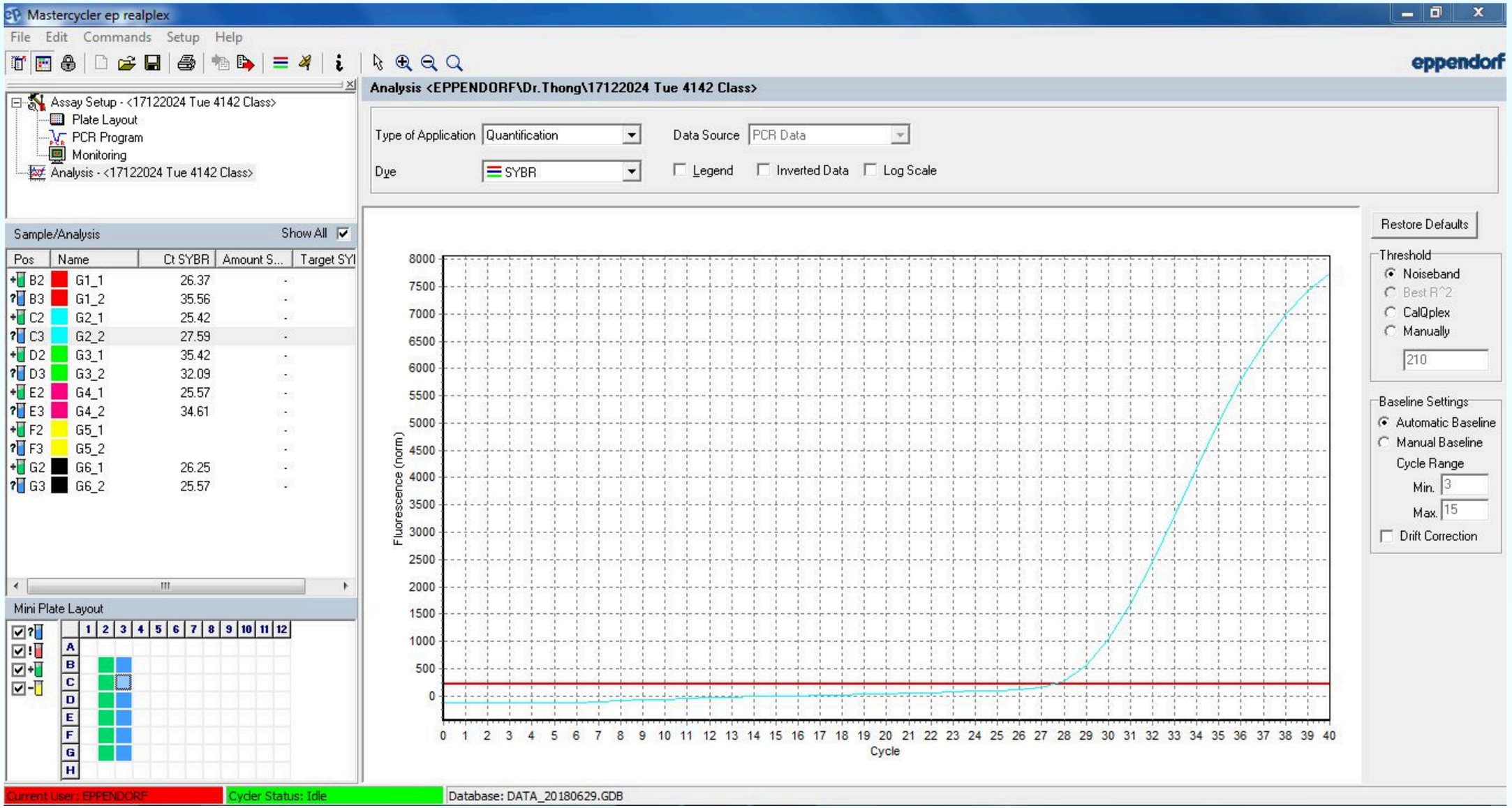
- Baseline: ~1-15th cycles, signal below background noise
- Exponential: ~15-30th cycles, products doubles each cycle
- Linear: reagent start to limit
- Plateau: late, dNTPs/primers depleted, Taq weakened



- **Baseline**, background in early cycles
- **Threshold**, line above baseline noise
- **Ct value**, the cycle at which a fluorescence first rises above background

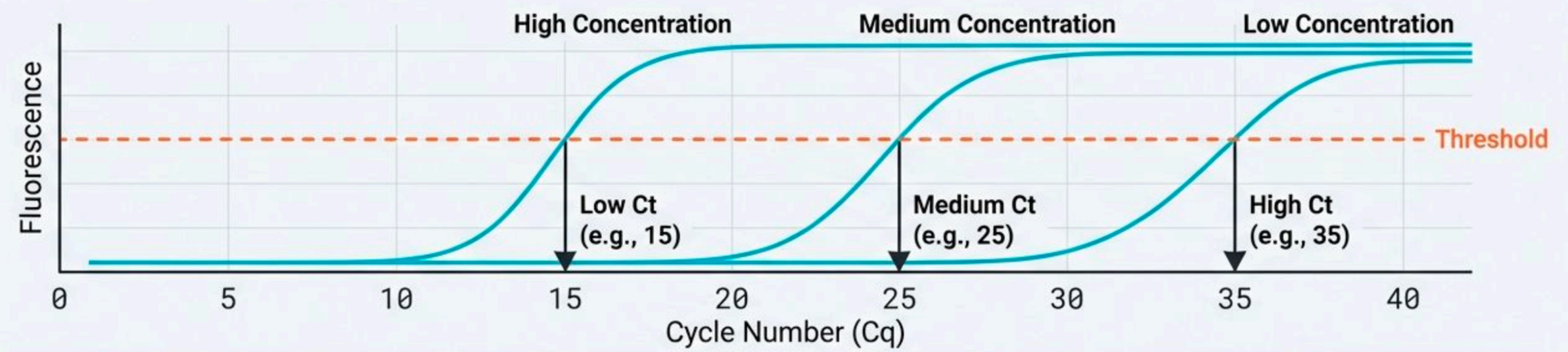


Ct value Qualitative - The target present or not?



Ct value Relative Quantification

Inverse rule: More template => Lower Ct

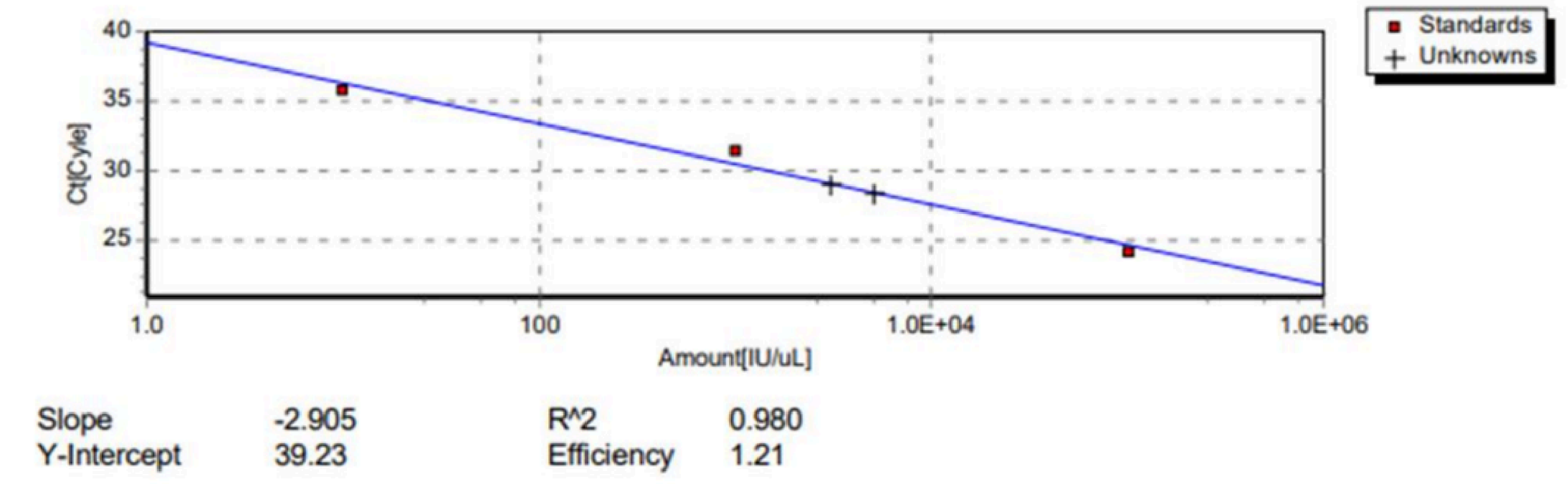


Ct is inversely proportional to the log of the initial template copy number.

a 10-fold difference = 3.32 cycles (2^n rule with n is no. of cycles)

Ct value Absolute Quantification

Standard curve

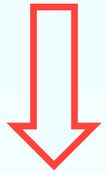


Ct values and the amount FAM of samples with the standards

Pos	Name	Ct FAM	Amount FAM [IU/uL]	Target FAM
B2	NC	-	-	NC
B3	E1	35.85	10.0	E1
B4	E3	31.49	1000	E3
B5	E5	24.23	1.00E+05	E5
B6	G1	29.12	3029	G1
B7	G2		-	G2
B8	G3	28.48	5049	G3
B9	G4		-	G4

HBV Standard:

- E1: 10^1 IU/uL
- E2: 10^2 IU/uL
- E3: 10^3 IU/uL
- E4: 10^4 IU/uL
- E5: 10^5 IU/uL

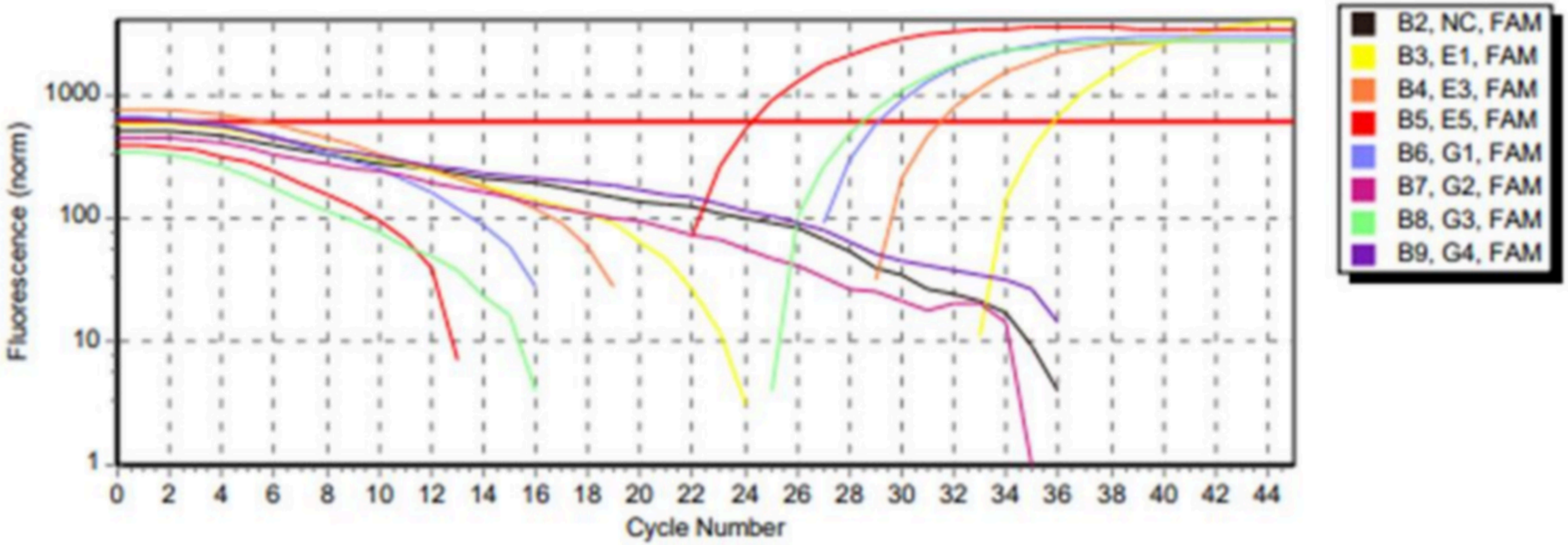


Sample: Viral load?

Infectious status?

Amplification Plot

Pos	Name	Ct FAM	Amount FAM [IU/uL]	Target FAM
B2	NC	-	-	NC
B3	E1	35.85	10.0	E1
B4	E3	31.49	1000	E3
B5	E5	24.23	1.00E+05	E5
B6	G1	29.12	3029	G1
B7	G2		-	G2
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B9	G4		-	G4

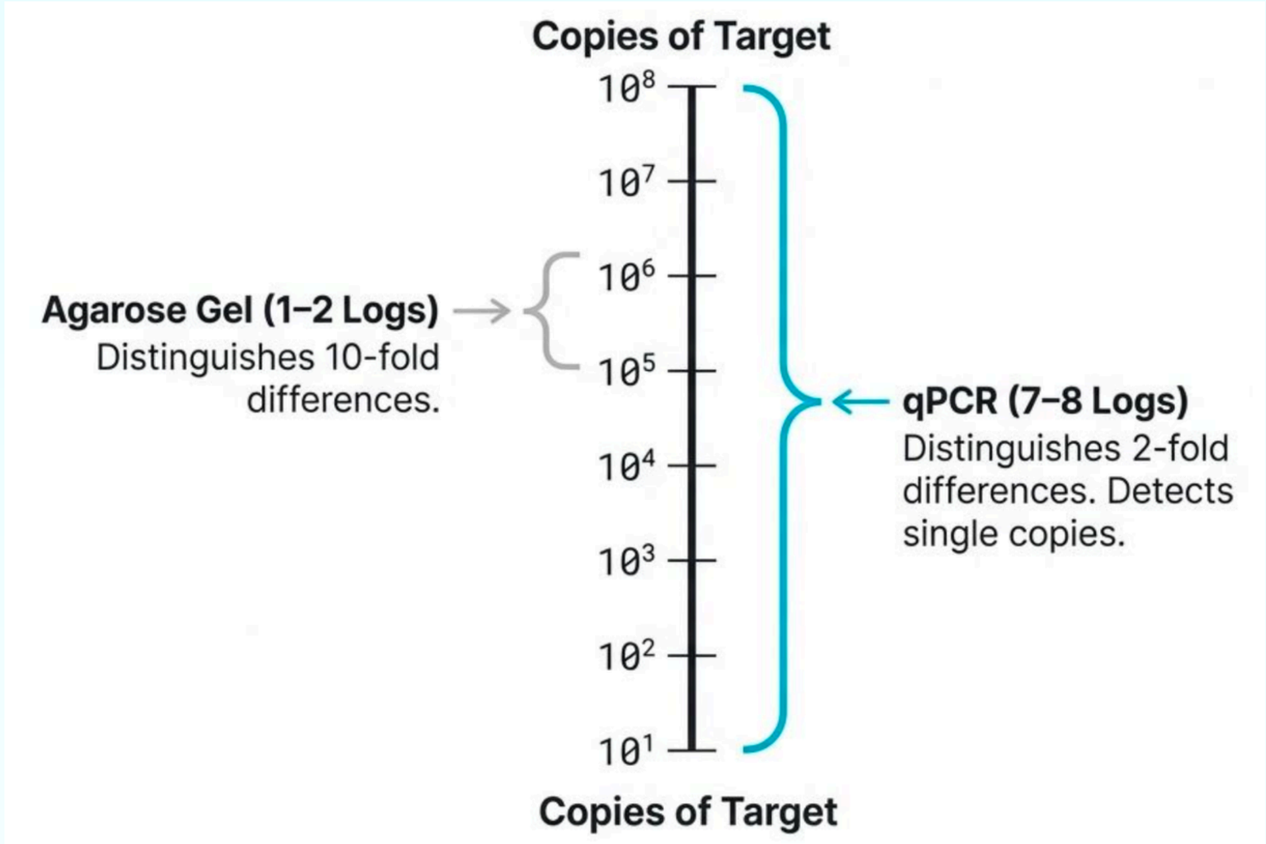


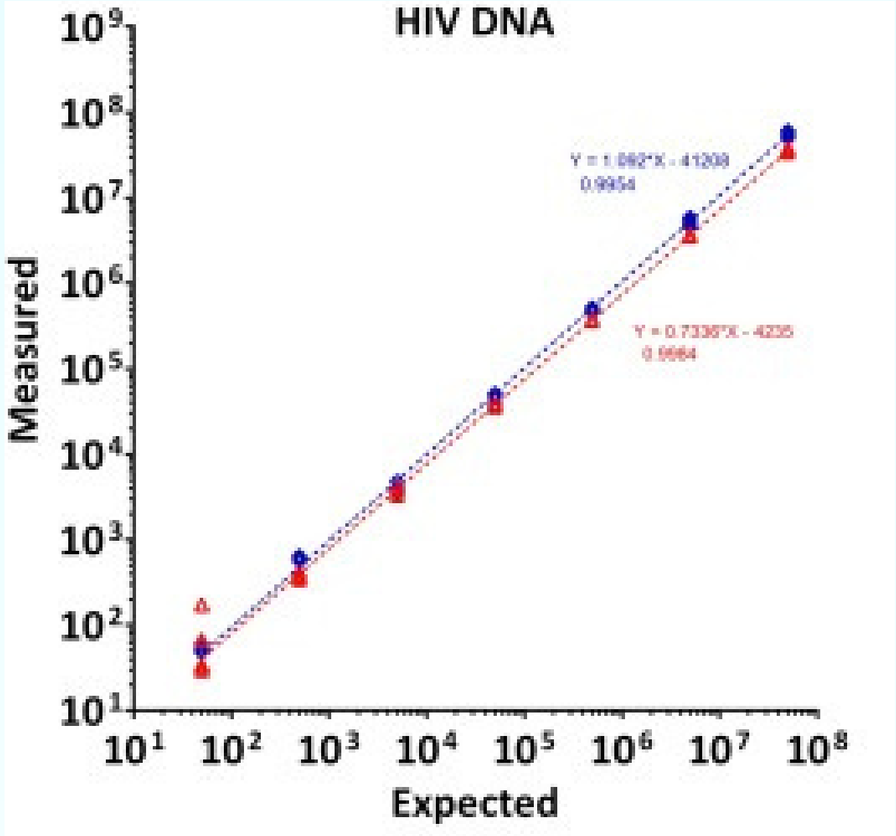
Threshold 612 (Noiseband)

Baseline automatic, Drift correction OFF

Dynamic Range

qPCR is linear across 7-8 orders of magnitude, versus 1-2 for gel band intensity





Ct value

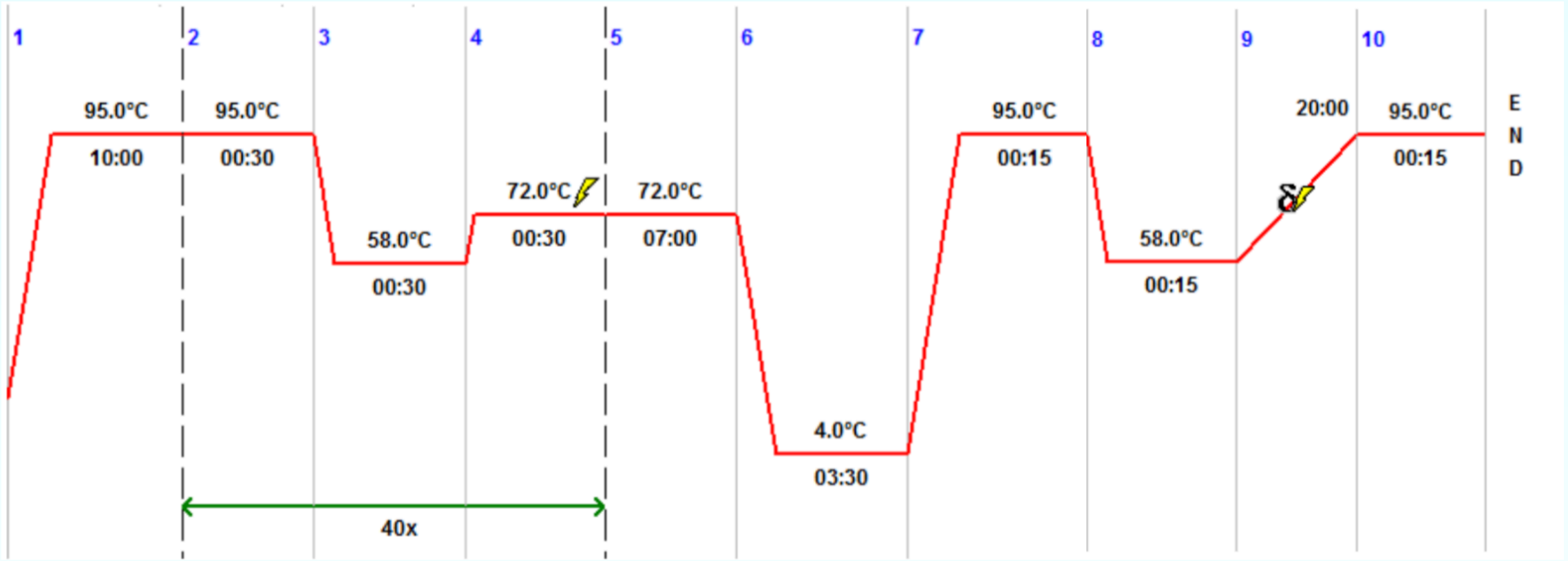
Qualitative

Related Quantitative

Absolute Quantitative

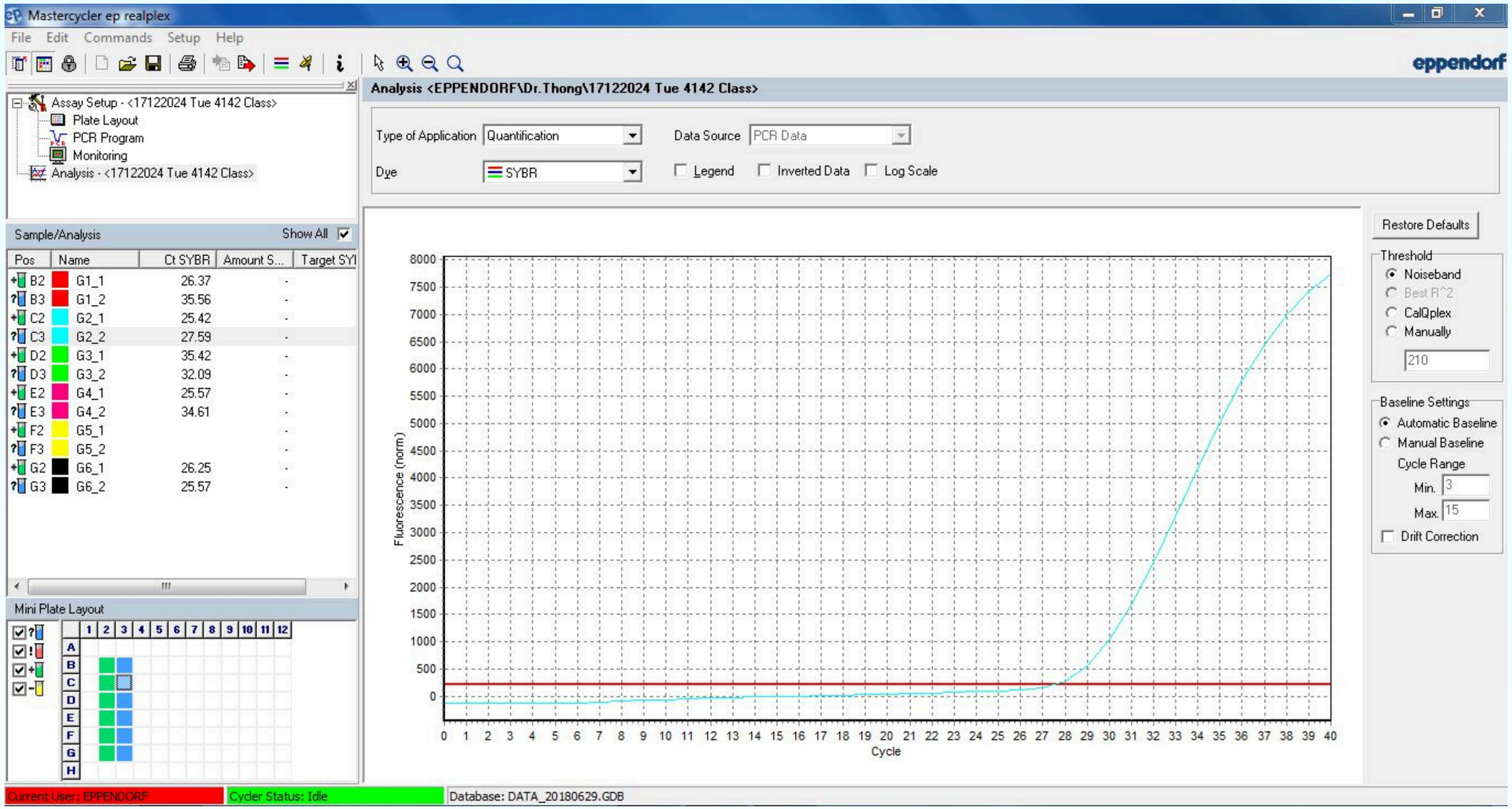
qPCR Reaction Protocol - Wetlab Exp.

Chemicals	(-) Control	(+) Control	Sample
DNA Template	0	1	1
F primer	1	1	1
R primer	1	1	1
Master Mix	10	10	10
Fluorescent dye	1	1	1
MiliQ	7	6	6
Total	20	20	20


1
2
3
4
5
6
7
8
9
10
95.0°C
10:00
95.0°C
00:30
58.0°C
00:30
72.0°C
00:30
72.0°C
07:00
4.0°C
03:30
95.0°C
00:15
58.0°C
00:15
20:00
95.0°C
00:15
E
N
D
40x

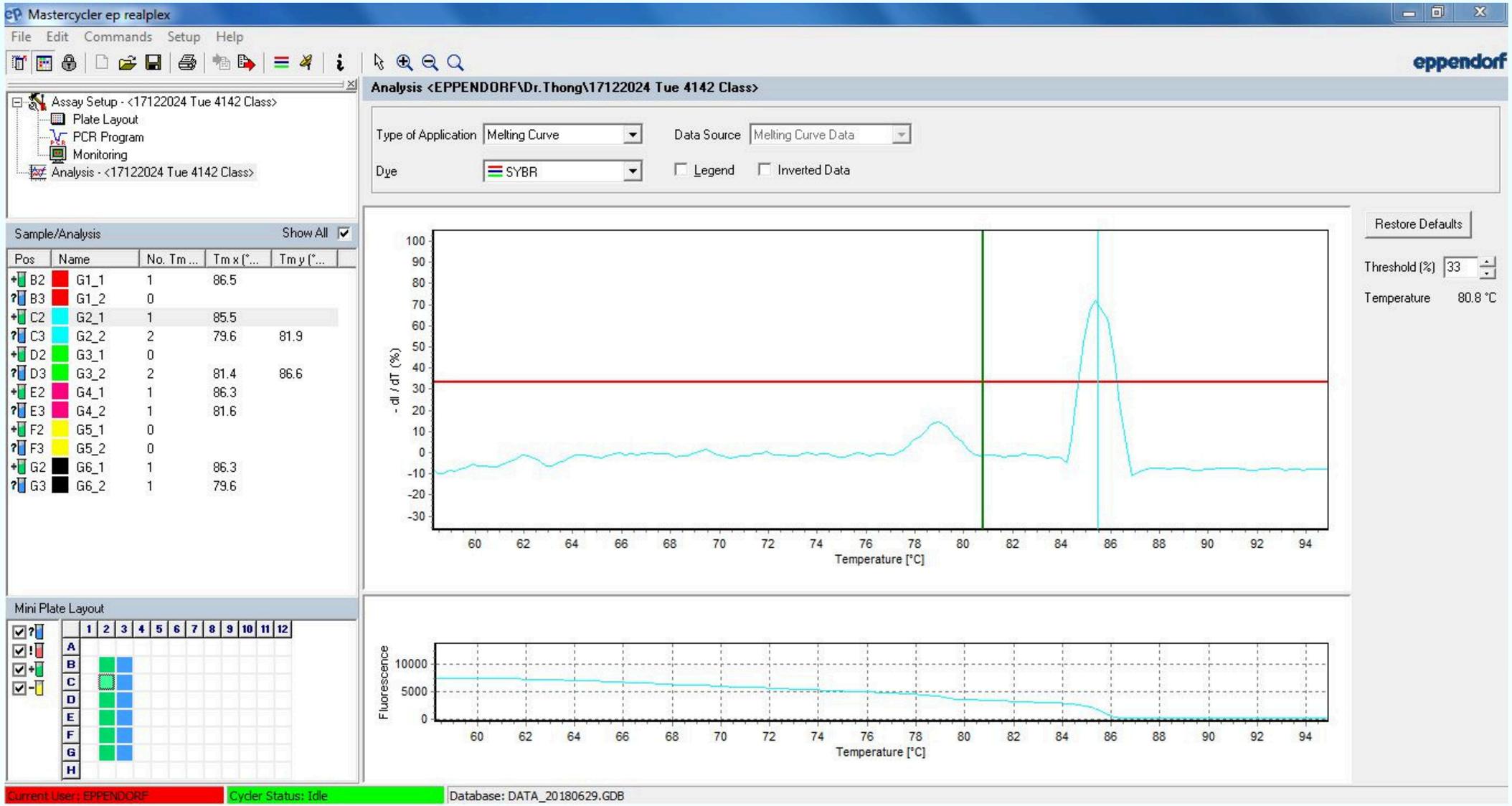
Beta-Thalassemia Detection by Realtime PCR

Ct value



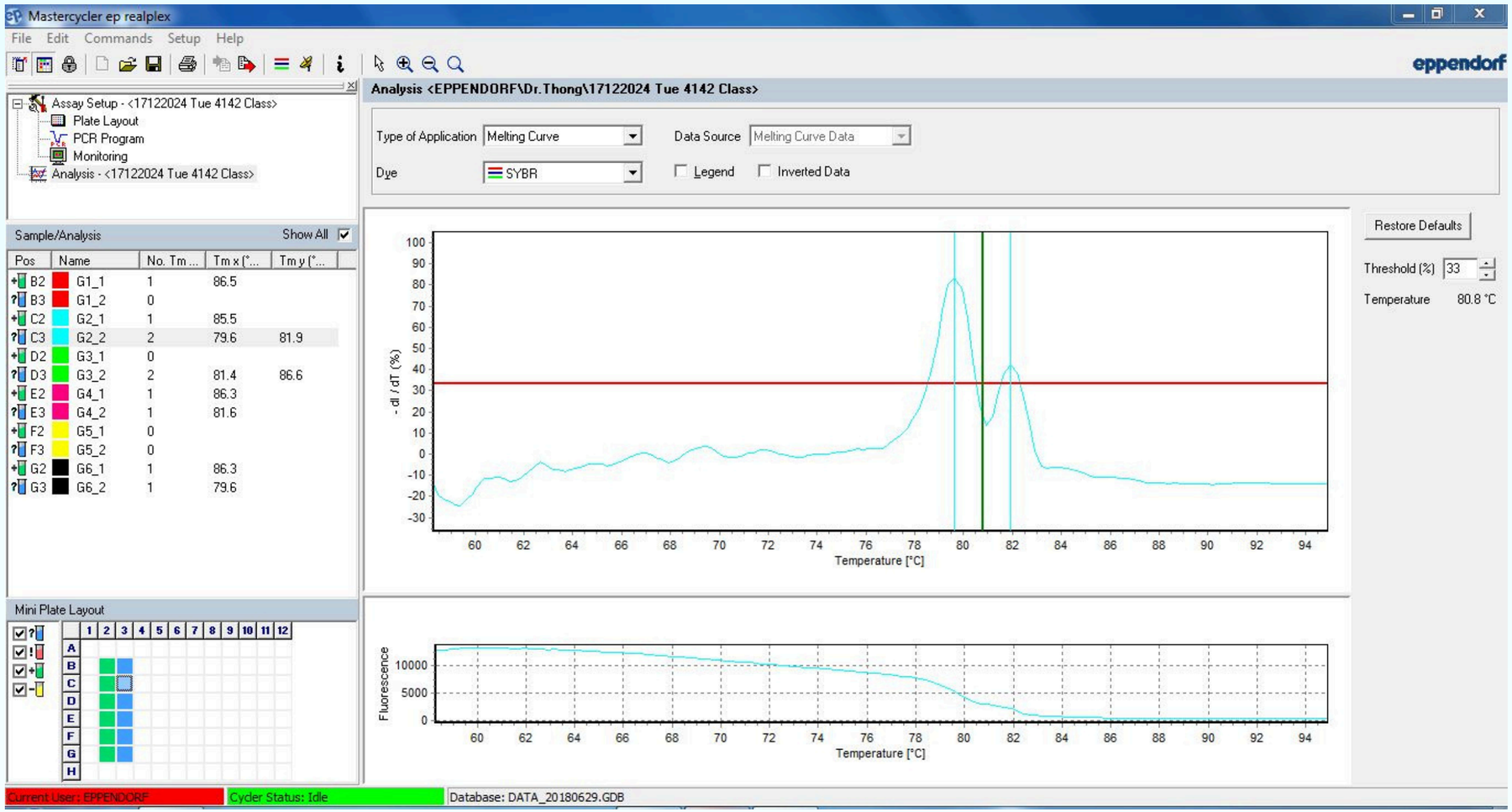
Beta-Thalassemia Detection by Realtime PCR

Positive Control



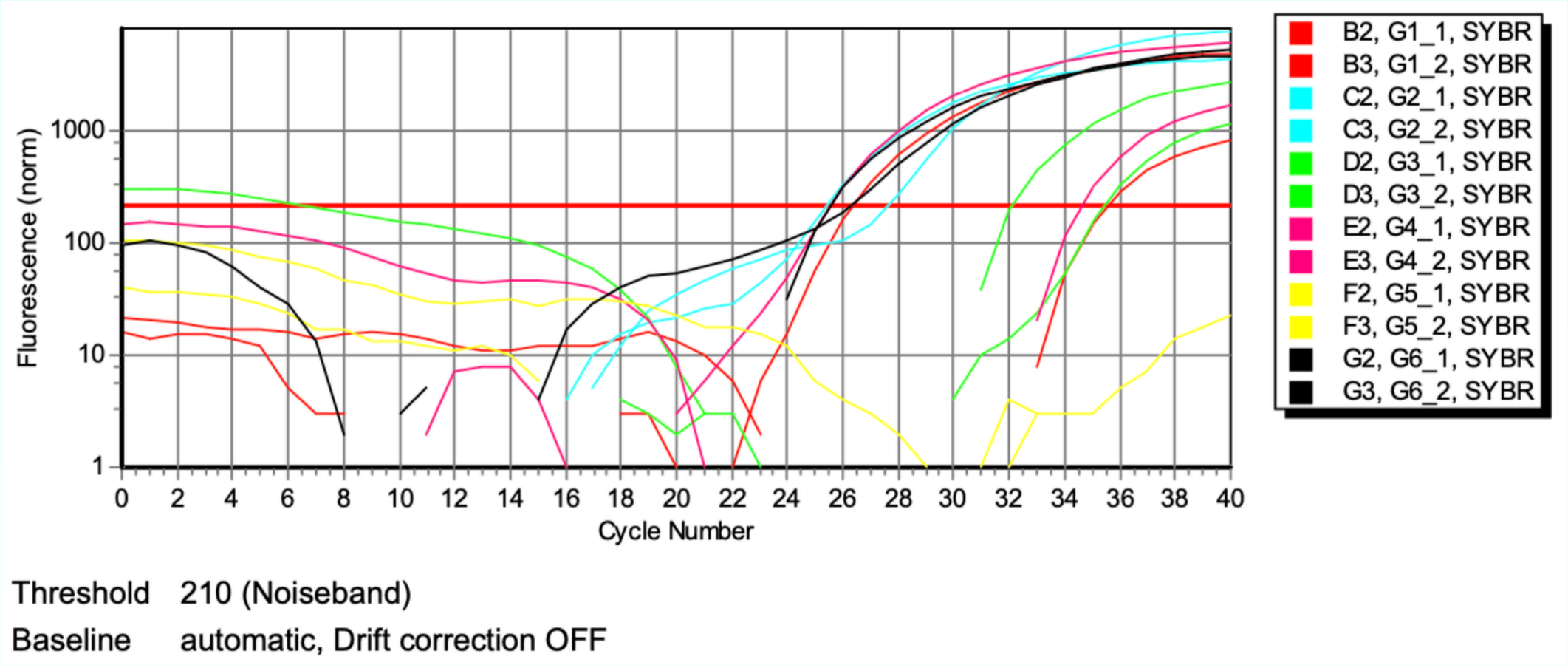
Beta-Thalassemia Detection by Realtime PCR

Sample



Beta-Thalassemia Detection by Realtime PCR

Amplification Plot



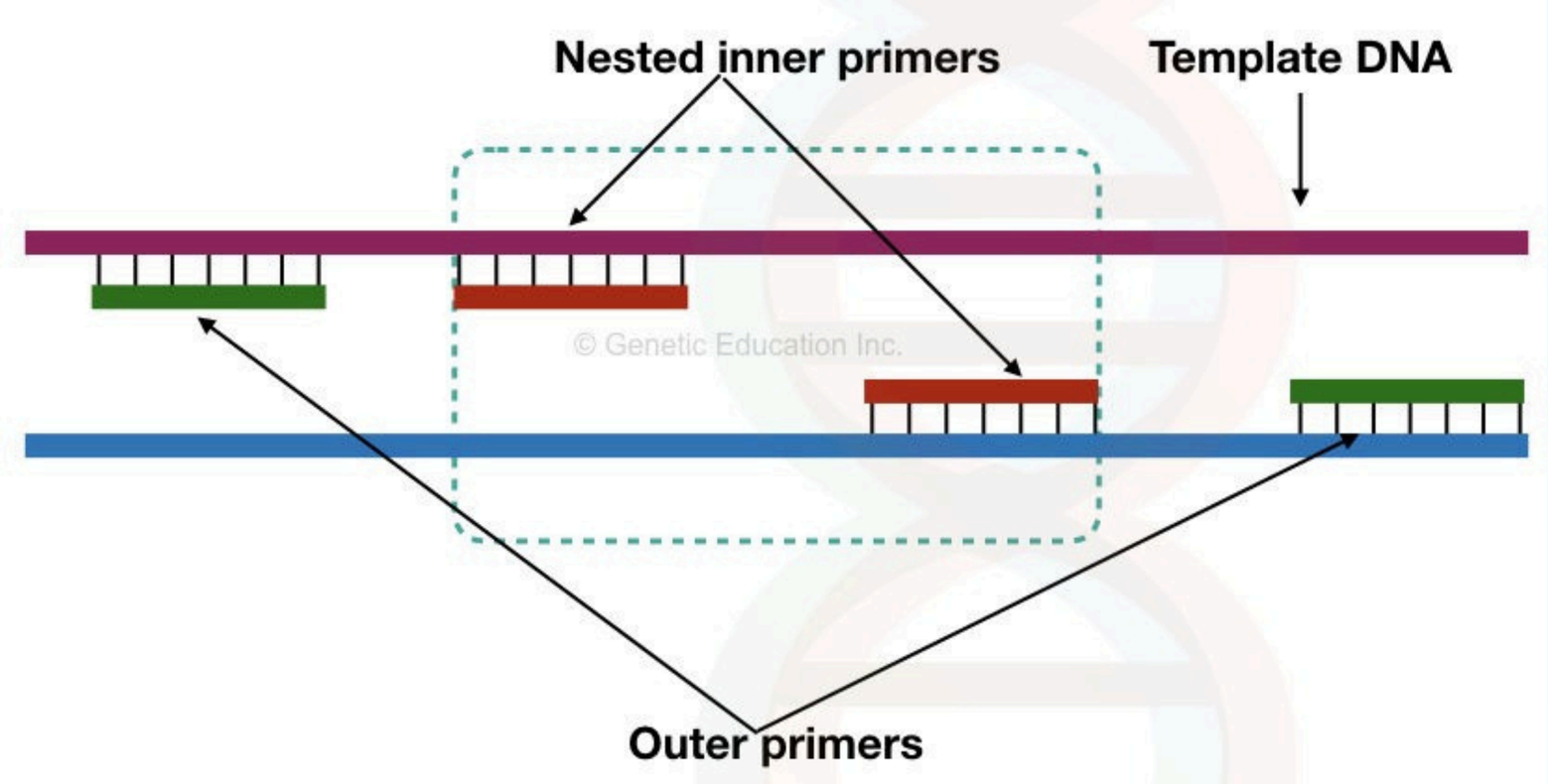
Threshold 210 (Noiseband)

Baseline automatic, Drift correction OFF

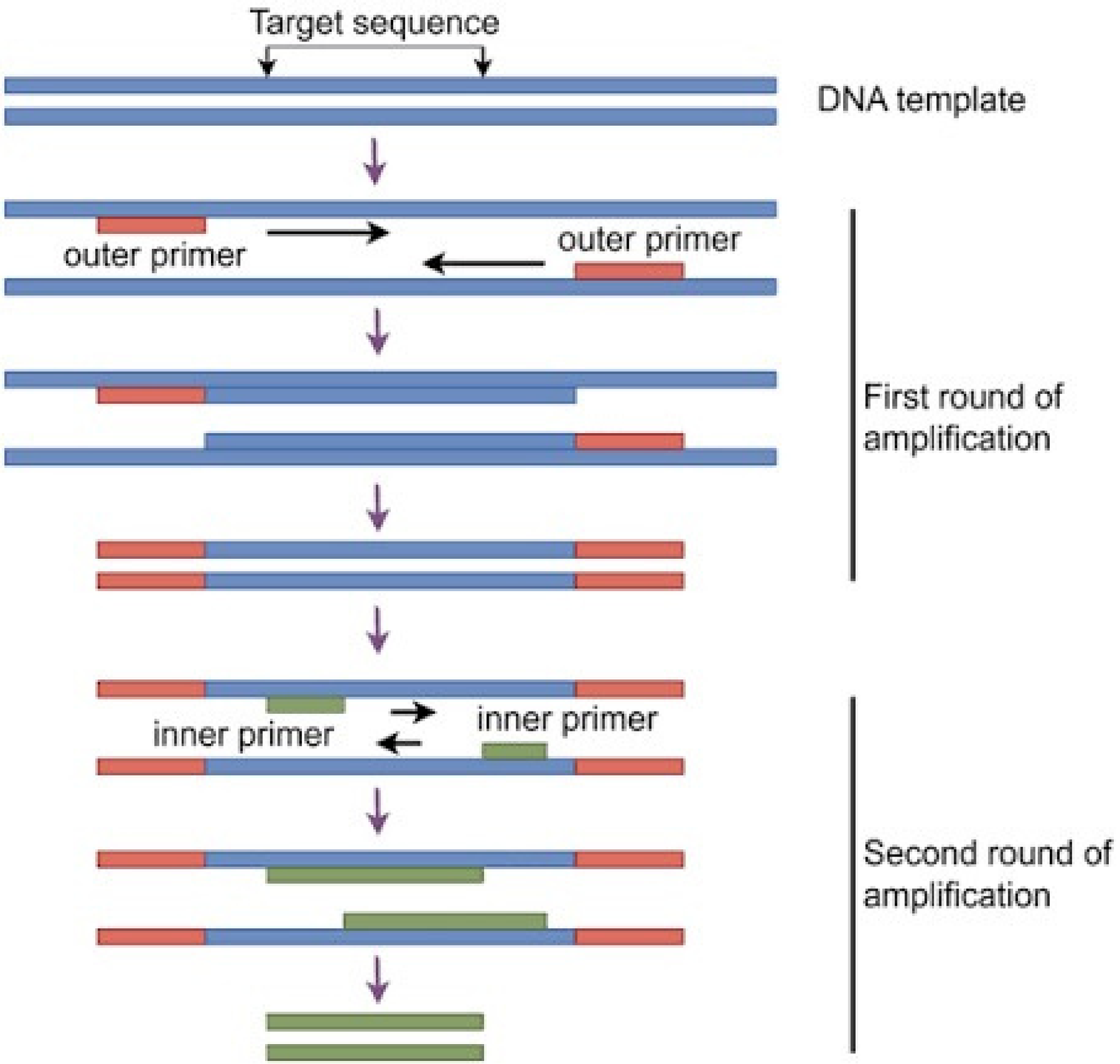
NESTED PCR

Nested PCR runs two PCRs with four primers

- Inner primer pair inside the first product



Why Nested PCR ??



Why it is more specific ?

Why it is more sensitive ?

Optimisation

- Outer primer are designed with higher Tm than inner
- Round 1 short, ~15-25 cycles => non-specific products and primer dimers do not accumulate before transfer
- Dilute first round products
- Titrate MgCl2 and primer concentration

Advantages

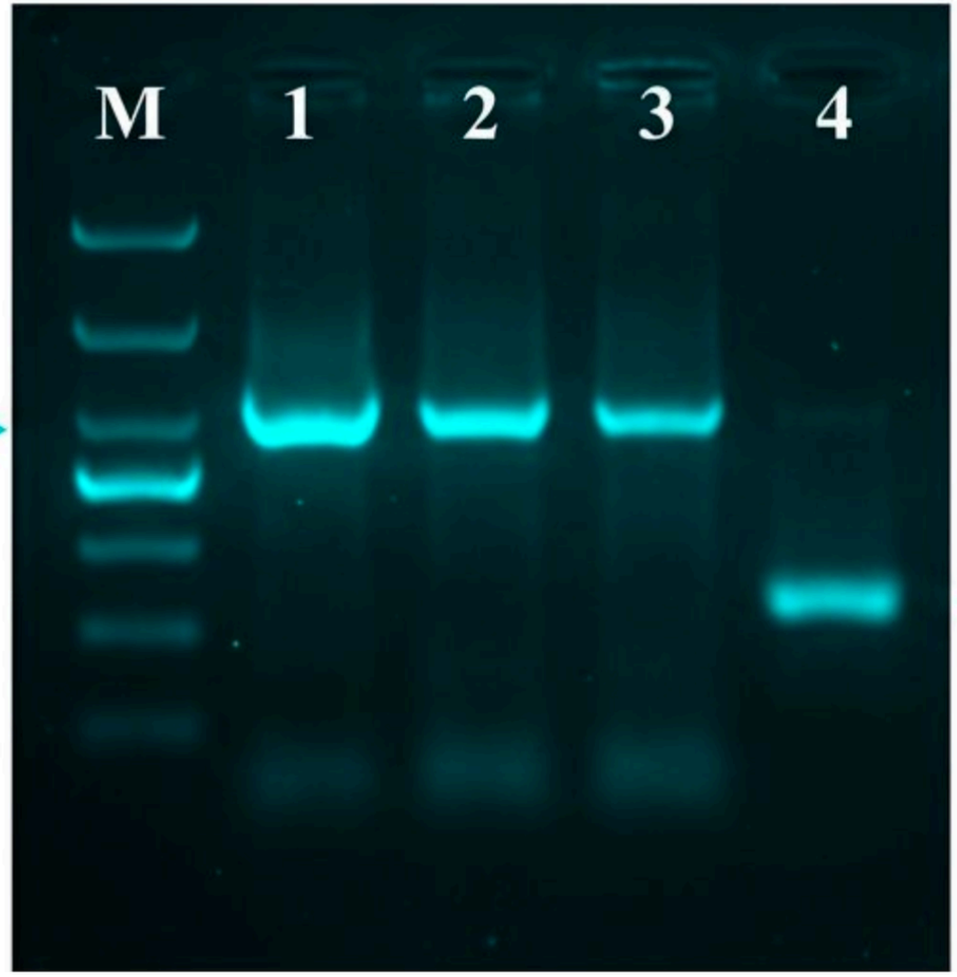
- The present technique is highly specific, sensitive and accurate;
- high amount of amplicons with greater specificity;
- offers amazing results for difficult templates like high GC-rich or longer templates;
- It’s useful for the amplification of low-abundance genes, alleles or DNA sequences.

Limitations

- Contamination is the main risk;
- Take a long time;
- Qualitative only;
- Requires an additional primer set and separate reaction setup which is tedious, and costly.

Aquaculture: Tracking *Metschnikowia bicuspidata* in Crabs

Used to discover and block transmission routes in reservoir hosts.



Outer Product: 493 bp

Inner Product: 226 bp

Procedure

Chemicals	(-) Control	(+) Control	Sample
DNA Template	0	1	1
Outer F primer	1	1	1
Outer R primer	1	1	1
Master Mix	10	10	10
Fluorescent dye	1	1	1
MiliQ	7	6	6
Total	20	20	20

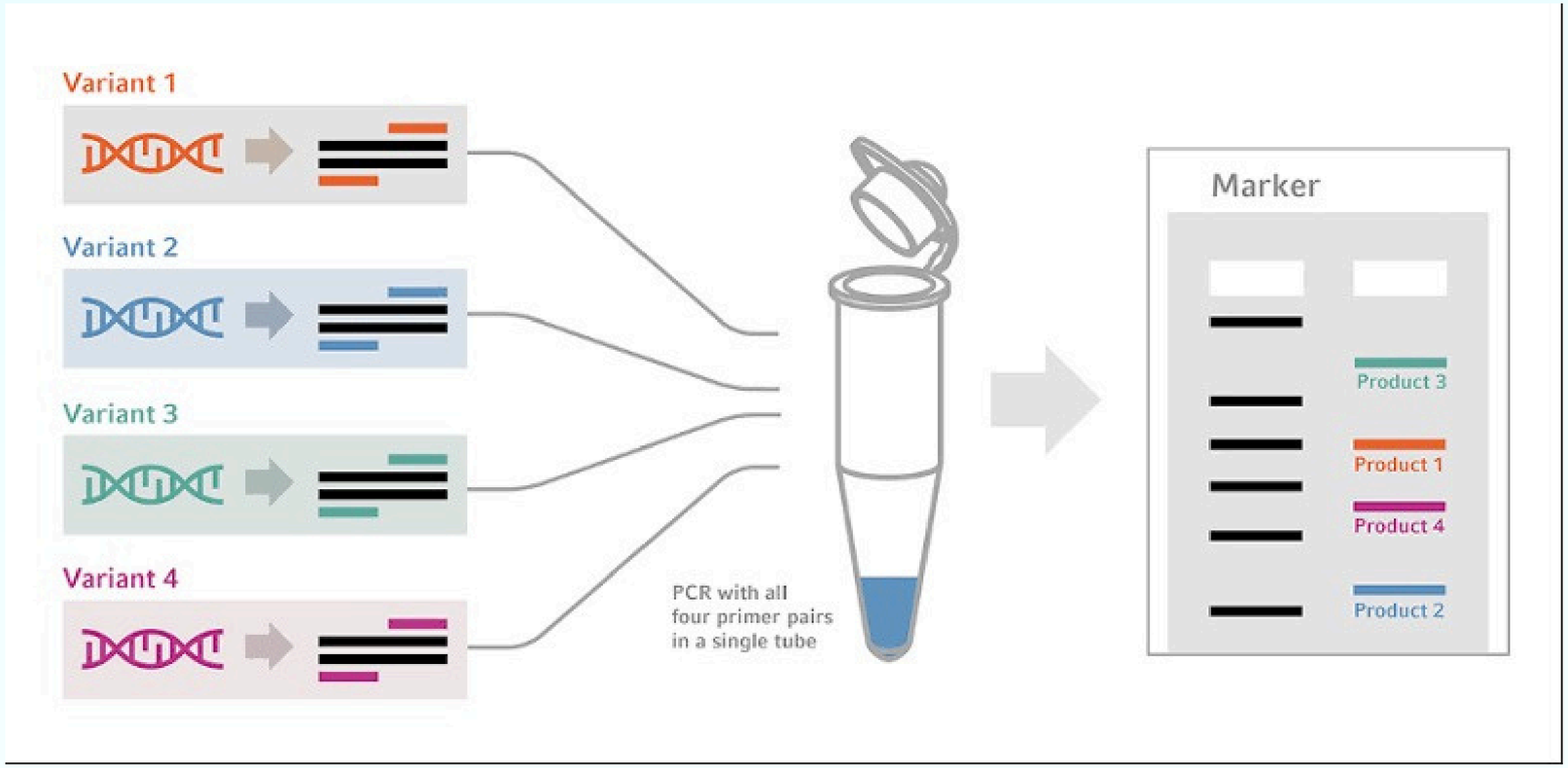
1st round

Chemicals	(-) Control	(+) Control	Sample
DNA Template	0	1	1
Inner F primer	1	1	1
Inner R primer	1	1	1
Master Mix	10	10	10
Fluorescent dye	1	1	1
MiliQ	7	6	6
Total	20	20	20

2nd round

MULTIPLEX PCR

- High-throughput detection principles;
- Multiplex PCR amplifies two or more targets with several primer pairs;
- Hardly keeping all targets amplifying evenly while they compete for the same Taq and dNTPs.



Why use it ??

Design rules

- All primers should fall in a narrow Tm window (~2-3°C)
- Distinct amplicon sizes
- No cross-complementary

Why targets drop out: PCR bias

PCR selection (deterministic)

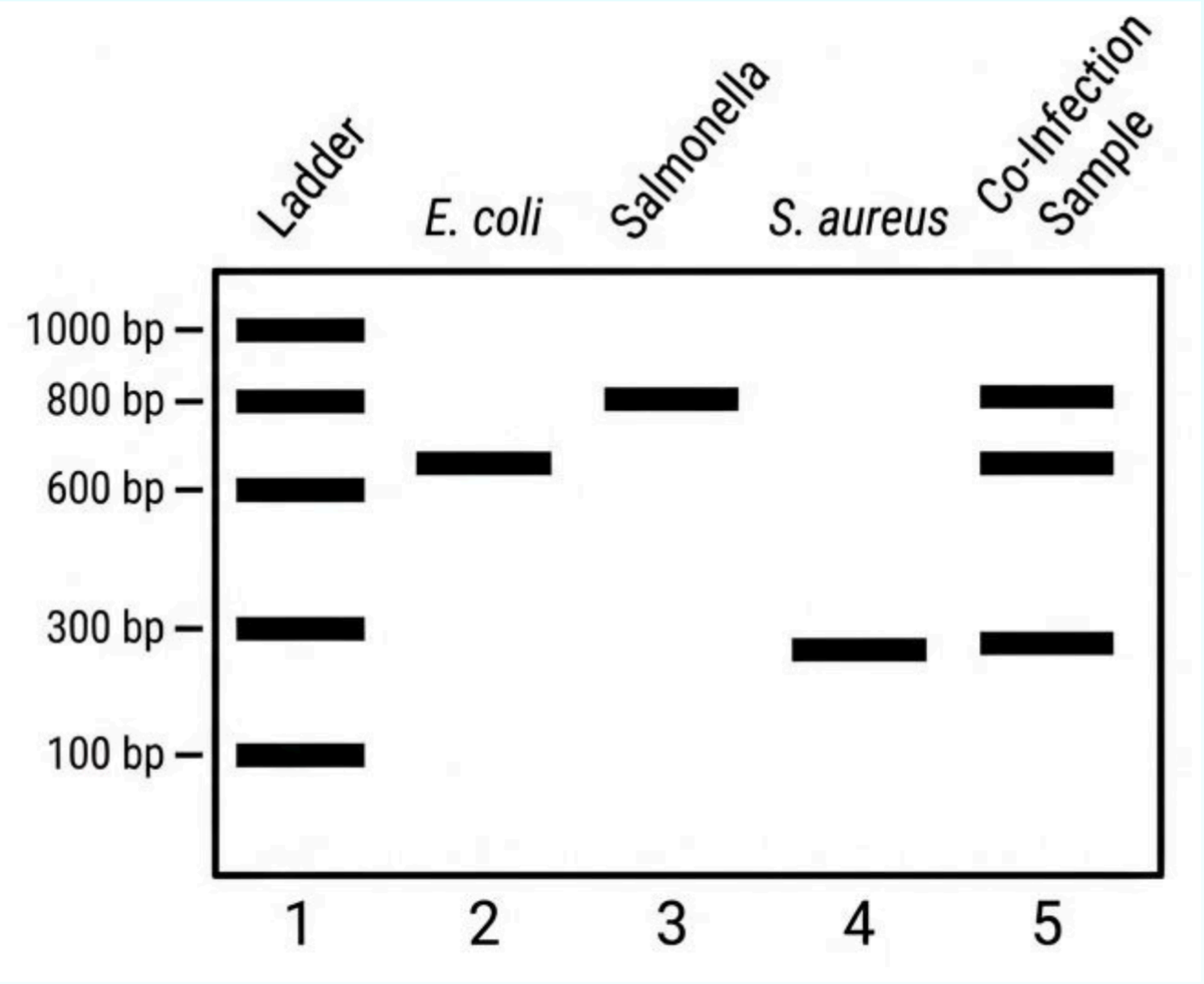
- Short amplicons amplify faster.
- High copy number templates dominate.

PCR drift (stochastic)

- Random fluctuations in early cycles.
- Critical impact on low-concentration samples.

Procedure

Chemicals	(-) Control	Sample 1	Sample 2
DNA Template	0	1	1
1st F-R primer	2	2	2
2nd F-R primer	2	2	2
3rd F-R primer	2	2	2
Master Mix	10	10	10
Fluorescent dye	1	1	1
MiliQ	3	2	2
Total	20	20	20


Ladder
E. coli
Salmonella
S. aureus
Co-Infection Sample
1000 bp
800 bp
600 bp
300 bp
100 bp
1
2
3
4
5

GAP PCR

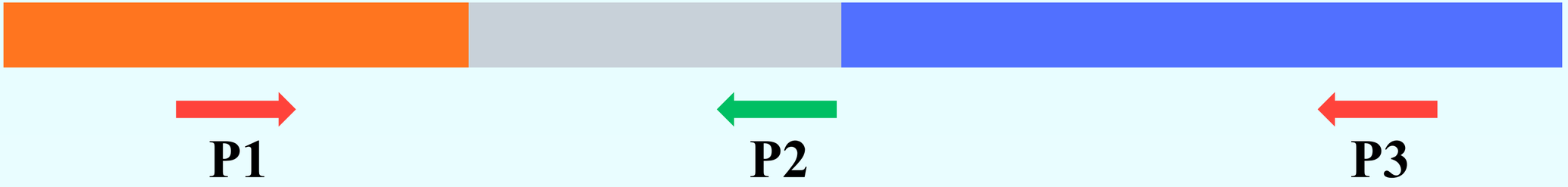
Gap-PCR detects large deletions with known breakpoints

Why standard PCR fails ??

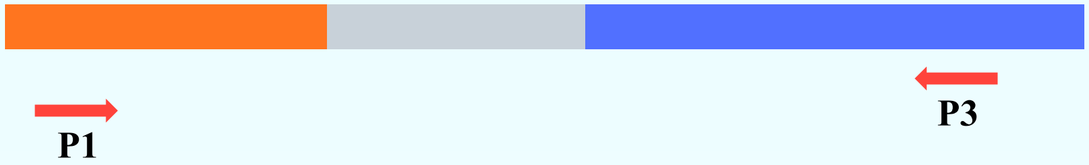
Principle

P1 (forward) and P3 (reverse) sit on either side of the deletion.

- Normal allele: P1–P3 distance > 20 kb → too long for Taq in a standard reaction → no band.
- Deleted allele: the gap is gone, P1–P3 now < 2 kb → band. *The band appears only because of the deletion.*

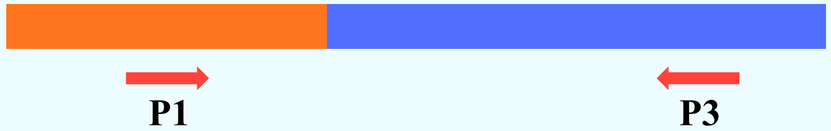


Normal Allele



no band

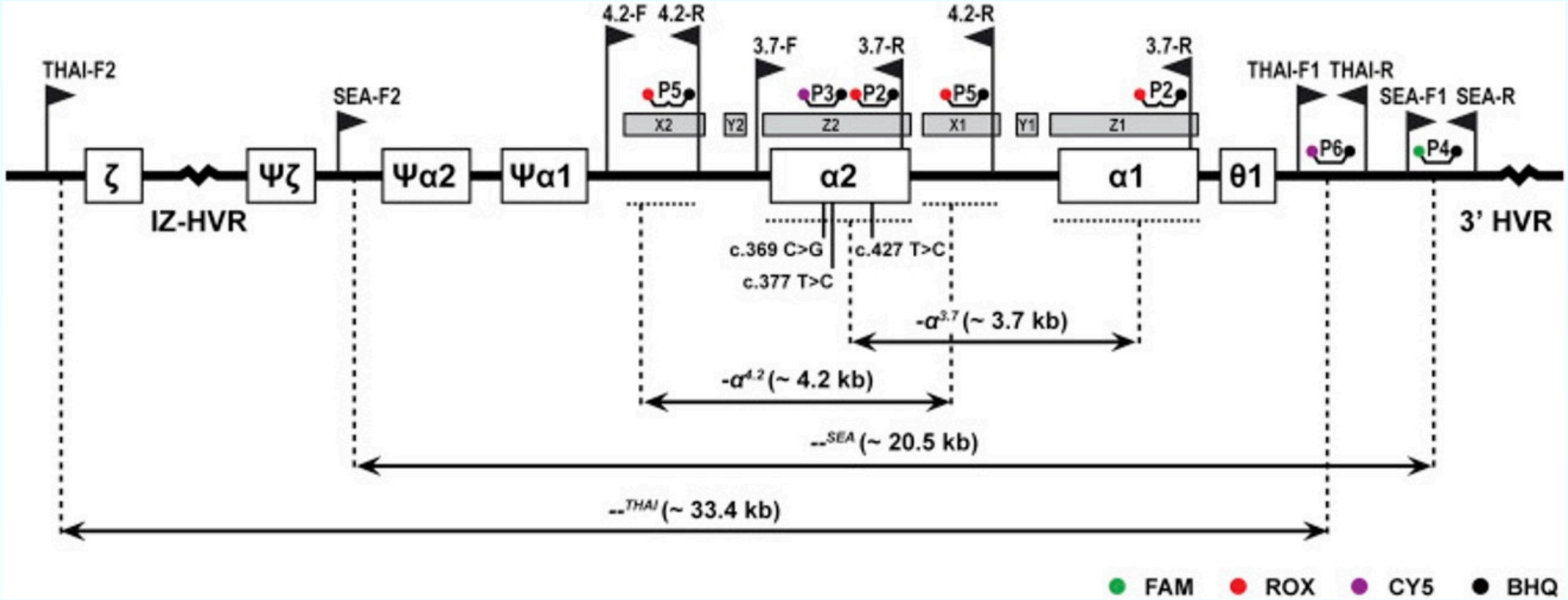
Deleted Allele



one band

Genotype	P1 + P2 (normal band)	P1 + P3 (deletion band)	
Normal (AA)	Y	N (too far)	1 band: normal
Heterozygous (Aa)	Y	Y	2 bands
Homozygous Deletion (aa)	N (P2 site deleted)	Y	1 band: deletion

Mutations of alpha thalassemia



PCR conditions

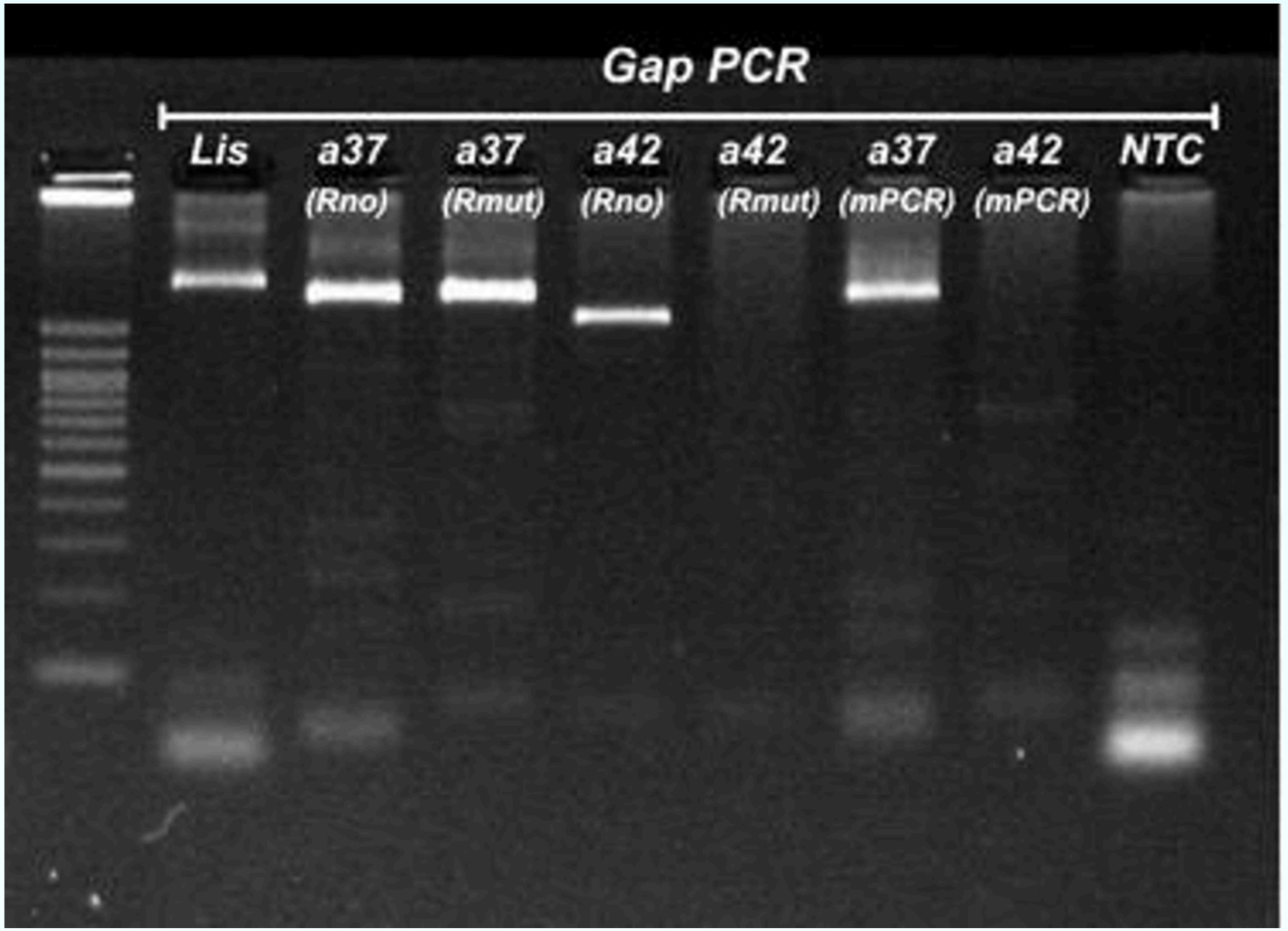
95oC 5 min

And 35 cycles of:

- 95oC for 30 sec,
- 60oC for 30 sec,
- 72oC for 2 min 30 sec,

72oC for 5 min

Hold at 4oC.


Gap PCR
Lis
a37 (Rno)
a37 (Rmut)
a42 (Rno)
a42 (Rmut)
a37 (mPCR)
a42 (mPCR)
NTC